



End Term (Odd) Semester Examination December 2024

Roll no. 2292104

Name of the Course and semester: **BCA 5th Sem**

Name of the Paper: *Programming with .NET C#*

Paper Code: *TBC-502*

Time: 3 hour

Maximum Marks: 100

Note:

- (i) All the questions are compulsory.
- (ii) Answer any two sub questions from a, b and c in each main question.
- (iii) Total marks for each question is 20 (twenty).
- (iv) Each sub-question carries 10 marks.

Q1. (2X10=20 Marks)

a. Explain the various components of NET framework with diagram. Also list the different languages supported by .Net Framework and explain the various services provided by CLR. CO1

b. Explain **assemblies** and different types of assemblies? Also explain .dll and an .exe file. CO1

c. Discuss the features and compilation process of .net framework? CO1

Q2. (2X10=20 Marks)

a. Explain the various methods of array class with proper syntax and example? Write the C# program to demonstrate the use of five pre-defined methods of array class. CO2

b. Write a menu driven program to show the use of five pre-defined methods of String class? CO2

c. Explain different types of array in C# with example? Write a C# program for the multiplication of 2-D matrix? CO2

Q3. (2X10=20 Marks)

a. Discuss the concept of inheritance in detail, also explain the types of inheritance with proper syntax and example. CO3

b. Describe the concept of function overloading and dynamic polymorphism?

Write a C# program to show the concept of virtual function and override keyword. CO3

c. Explain constructors and different types of constructors in C#. Also write a program to demonstrate the same. CO3

Q4. (2X10=20 Marks)

a. Discuss the concept of ASP.NET and describe the components of ASP.NET MVC? Explain its advantages. CO4

b. Explain ASP.NET with its features also explain Web forms and server controls. CO4

c. Explain the purpose of master pages in ASP.NET and how do they help in maintaining consistency across the website. CO4

Q5. (2X10=20 Marks)

a. Discuss the steps to establish a connection to a SQL server database using ADO.NET? CO5



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- b. Describe RDMS? Explain the different keys used in DBMS with example. CO5
- c. Discuss the architecture of ADO.NET? Explain the difference between ADO and ADO.NET? CO5